Nansen Converter

User guide

EAF-Nansen tool suite

20 October 2026

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1 Overview

Nansen Converter turns survey data into the two NMD XML formats that StoX reads: **NMDBiotic v3.1** for trawl stations, catches and individual fish, and **NMD Echosounder v1** for acoustic NASC. Each conversion takes two steps, with an Excel workbook in between that you can check and correct before the XML is written.

Table 1: The four converters

Step	Tab	Input	Output
1	BioData_To_Biotic.xls	Stations, catches, length-frequency and biology sheets	`Biotic.xlsx`
2	Biotic.xls_To_Biotic.xml	`Biotic.xlsx`	`Biotic.xml`
3	NASC.xls_To_Acoustic.xml	Echoview NASC export	`Acoustic.xlsx`
4	Acoustic.xls_To_Acoustic.xml	`Acoustic.xlsx`	`Acoustic_<date-time>.xml`

Three more tabs support them. **Code lists** chooses the codes used for every coded field, **Excel file templates** downloads example inputs, and **User_Guide** downloads this guide.

Both XML steps validate the data before you download it: keys and links between tables, codes, plausibility checks on stations, catches, fish and acoustic intervals, species names, the official NMD schema, and a read-back of the file with RstoxData. Errors block the download unless you allow it.

1.1 Starting the app

The app needs R 4.6 and the packages listed in the README. Two of them come from the EAF-Nansen tool suite on the IMR GitLab: **nansencore** (NMD reference codes, maritime zones) and **nansentaxonomy** (species names, aphia codes and acoustic categories).

Open `nansenconvert.Rproj` in RStudio and run `shiny::runApp()`, or run `Rscript -e "shiny::runApp()"` from the project folder. The app opens on a start screen (Figure 1); click **Launch the application**.

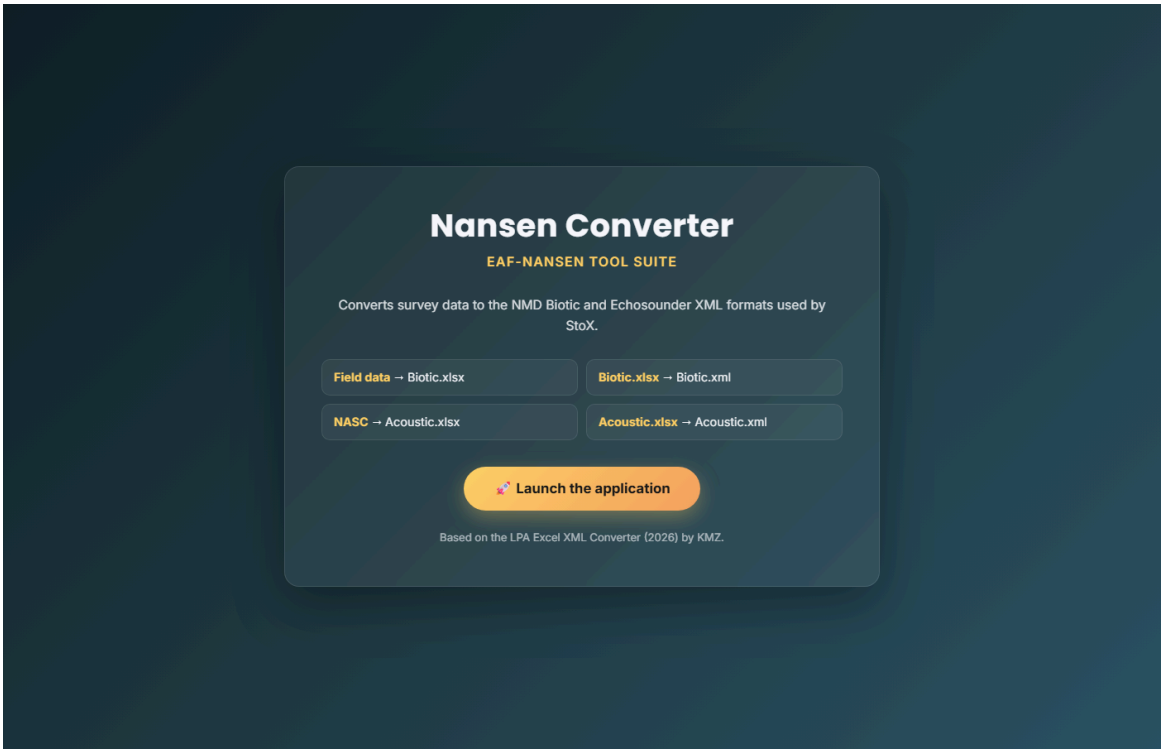


Figure 1: The start screen

1.2 Before you start

- **Start from the templates.** The *Excel file templates* tab has an example of every input, filled with a fictional survey. Their column names are the ones the converters look for (Section 7).
- **Choose your code lists first** if your data use codes other than the NMD reference ones (Section 6). The converters use whatever the Code lists tab holds at the time.
- **Regenerate old files.** Workbooks and XML made with the earlier LPA Excel XML Converter differ in sex codes, acoustic category codes, species NASC values, interval lengths, times and catch weights. Convert the original data again rather than reusing them.

2 Field data → Biotic workbook

The **BioData_To_Biotic.xls** tab has four sub-tabs, used in order. The first three each make one Excel file; the fourth combines them into `Biotic.xlsx`.

2.1 Fishstations

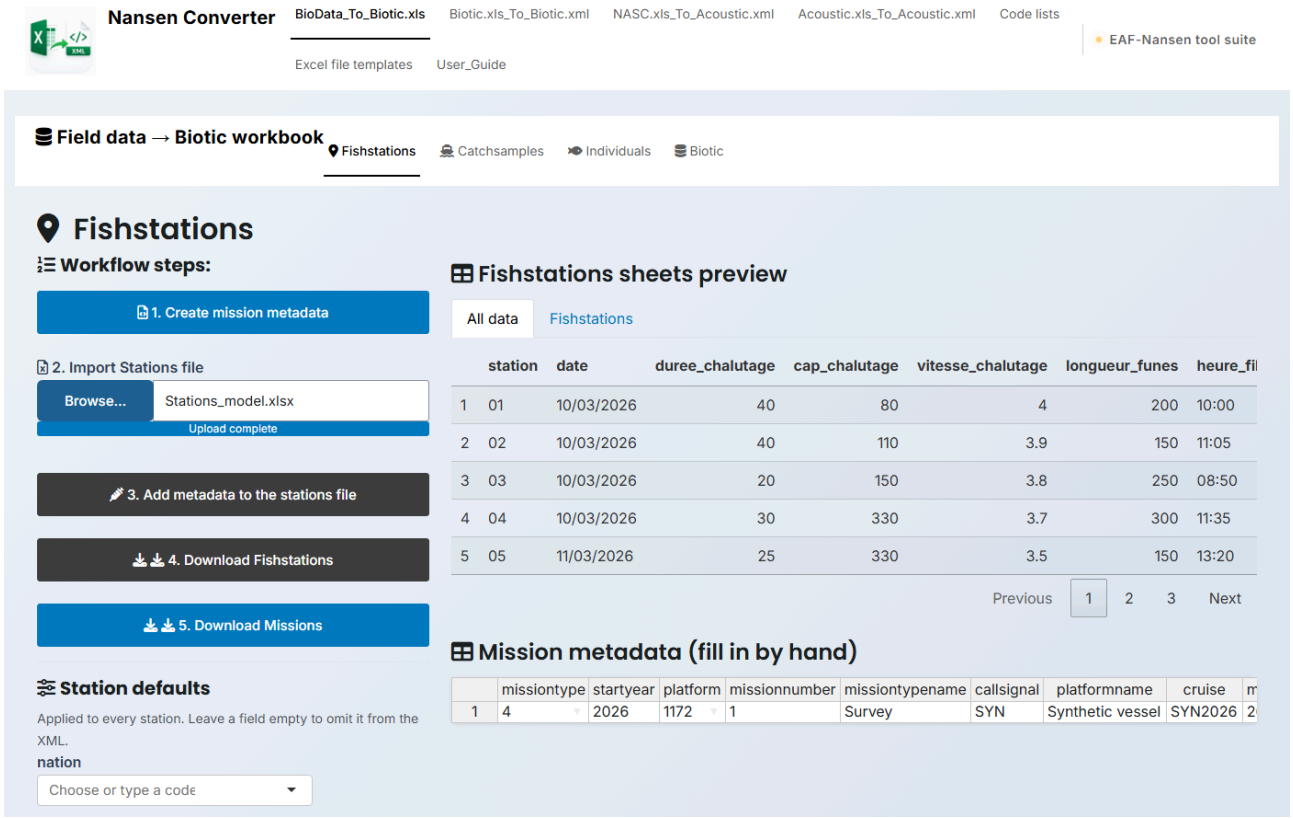


Figure 2: Fishstations: the workflow on the left, the stations preview and the mission metadata table on the right

Follow the numbered buttons (Figure 2):

1. **Create mission metadata.** A one-row table appears under the preview. Fill in `missiontype` and `platform` (both offer the codes of the active code lists), `startyear`, `missionnumber`, `missiontypename`, `callsignal`, `platformname`, `cruise`, `missionstartdate`, `missionstopdate` (for example `2026-03-10Z`) and `purpose`.
2. **Import Stations file** (`.xlsx`, `.xls` or `.csv`).
3. **Add metadata to the stations file.** This copies the mission number and platform to every station.
4. **Download Fishstations** and 5. **Download Missions**.

The **Station defaults** in the sidebar apply to every station: `nation`, `area`, `gear` (default 3500), `gearcount` (1), `haulvalidity` (1), `stationtype`, `trawldoorspread` (15 m) and `verticaltrawlopening` (50 m). Nation and area have no default; leave a field empty to leave it out of the XML.

Column names are matched without regard to case or accents (`Durée_chalutage` is read as `duree_chalutage`). The Stations file columns become these NMDBiotic fields:

Table 2: Stations columns and the fields they fill

Stations column	NMDBiotic field
`Station`	`serialnumber`
`Date`	`stationstartdate`, `stationstopdate`
`Heure_filage`, `Heure_virage`	`stationstarttime`, `stationstoptime`
`Latitude_filage`, `Longitude_filage`	`latitudestart`, `longitudestart`
`Latitude_virage`, `Longitude_virage`	`latitudeend`, `longitudeend`
`Prof_fond_filage`, `Prof_fond_virage`	`bottomdepthstart`, `bottomdepthstop`
`Prof_filet_filage`, `Prof_filet_virage`	`fishingdepthmin`, `fishingdepthmax`
`Durée_chalutage` (minutes)	`soaktime` (hours)
`Vitesse_chalutage`	`vesselspeed`
`Cap_chalutage`	`direction`
`Longueur_funes`	`wirelength`
`Note`	`stationcomment`

The towed `distance` (nautical miles) is computed from the start and end positions. Missing positions are highlighted in yellow in the preview, and you can edit any cell there. When a start or stop time is missing, it is filled in as 30 minutes before or after the other one.

2.2 Catchsamples

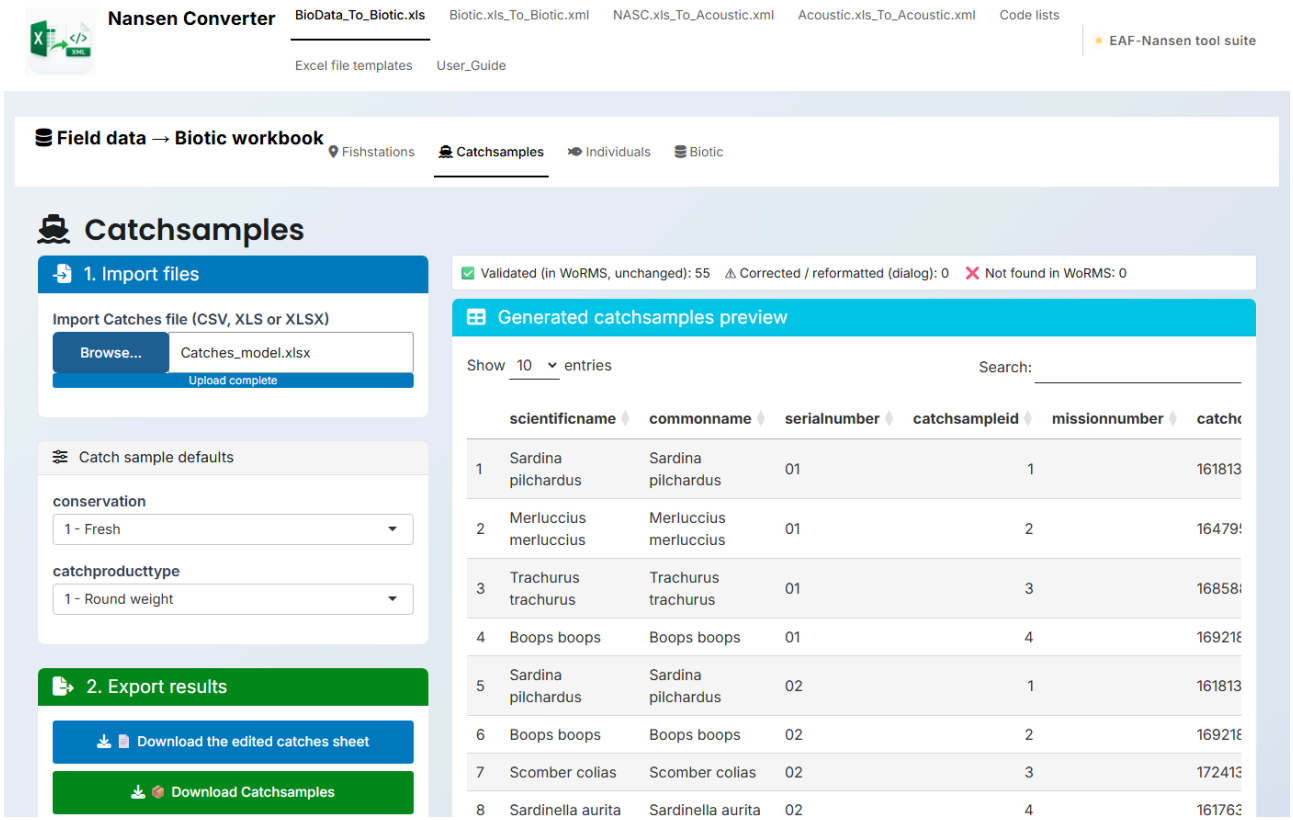


Figure 3: Catchsamples after import

Import the Catches file (`.csv`, `.xls` or `.xlsx`). The columns are found by name:

Table 3: Catches columns

Meaning	Column (accepted names)	
Species	`Especе` (also espèce, esp, scientificname, species)	required
Station	`Station` (also serial, serialnumber)	required
Sample weight, kg	`Ech_P`	required
Number in the catch	`Tot_N`	required
Total catch weight, kg	`Tot_P`	optional

`catchweight` is the whole catch (`Tot_P`, or `Ech_P` when there is no `Tot_P`), `catchcount` is `Tot_N`, and `raisingfactor` is `Tot_P / Ech_P`. The aphia code and `catchcategory` of each species come from nansentaxonomy. The sidebar sets `conservation` and `catchproducttype` (both 1 by default).

Species names. The names are checked against the EAF-Nansen taxonomy. If some are not found, a dialog lists them with the likely correct name pre-selected; type the start of a name to search the list. **Check these names in WoRMS** looks them up online, and names found only in WoRMS can be downloaded as a CSV for the next nansentaxonomy release. **Apply replacements** uses your choices; **Continue without corrections** keeps the imported names, which leaves those species without an aphia code (an error in the Biotic XML validation).

Download the result with **Download Catchsamples**. **Download the edited catches sheet** saves your input with the corrected names.

2.3 Individuals

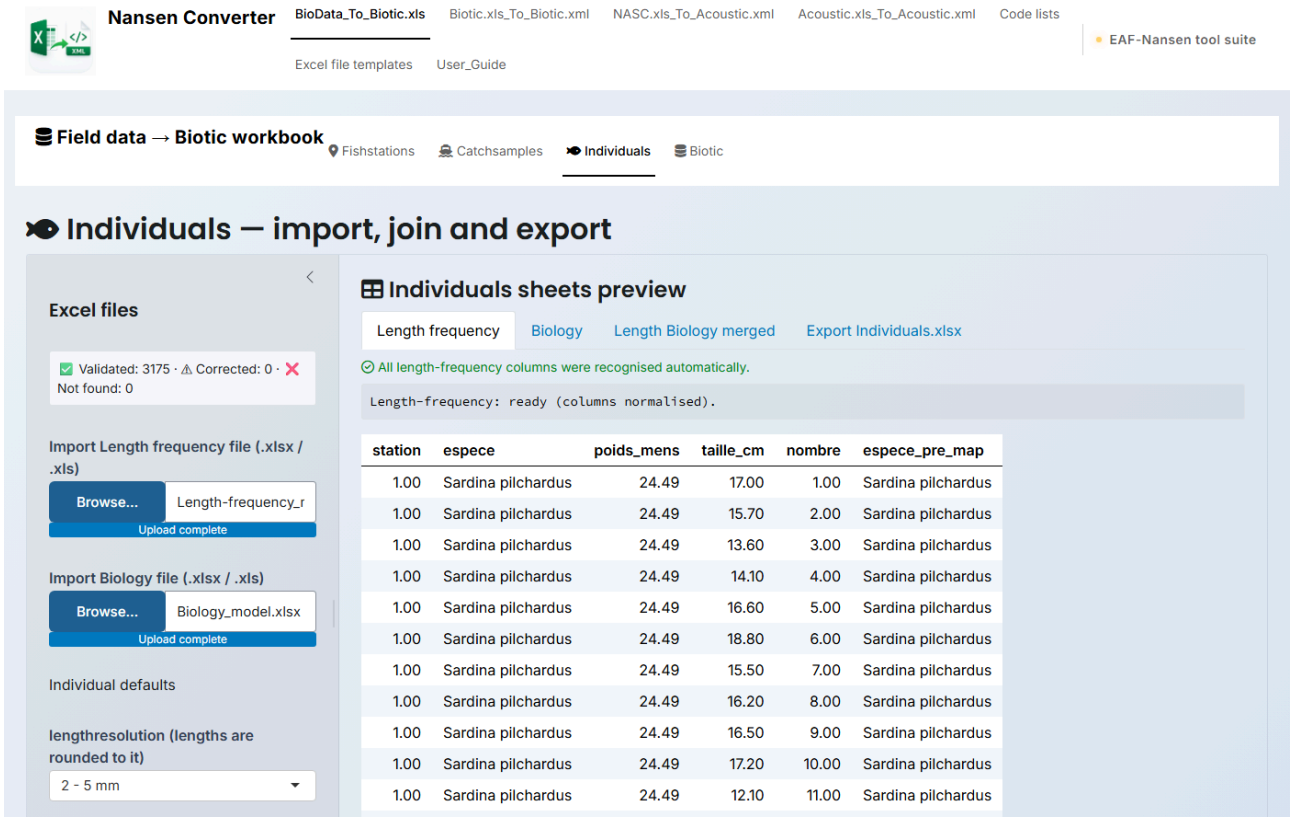


Figure 4: Individuals: the length-frequency and biology files joined

Import a **Length-frequency** file (`station`, `especе`, `Poids_Mens`, `taille_cm`, `nombre`) and a **Biology** file (`especе`, `station`, `nombre`, `poids_g`, `sexe`, `sms`; other columns are ignored). If a column has another name, choose it in the column-matching panel of the preview tabs. Species names are checked as in Catchsamples.

Join Biology with Length pairs the rows of the two files with the same station, species and `nombre`. When several rows share the same `nombre`, the first is paired with the first, the second with the second, and so on. Then:

- `taille_cm` becomes `length` in metres, rounded to the **lengthresolution** chosen in the sidebar (default 2, 5 mm);
- `poids_g` becomes `individualweight` in kilograms;
- `sexe` becomes `sex` through the Sex code list (F → 1 Female, M → 2 Male, I left empty);
- `sms` becomes `maturationstage` through the Maturation stage code list;
- `Poids_Mens` is kept as the weight of the length sample.

Export Individuals.xlsx saves the joined fish. The *Export Individuals.xlsx* preview tab shows the exported columns and their order.

2.4 Biotic

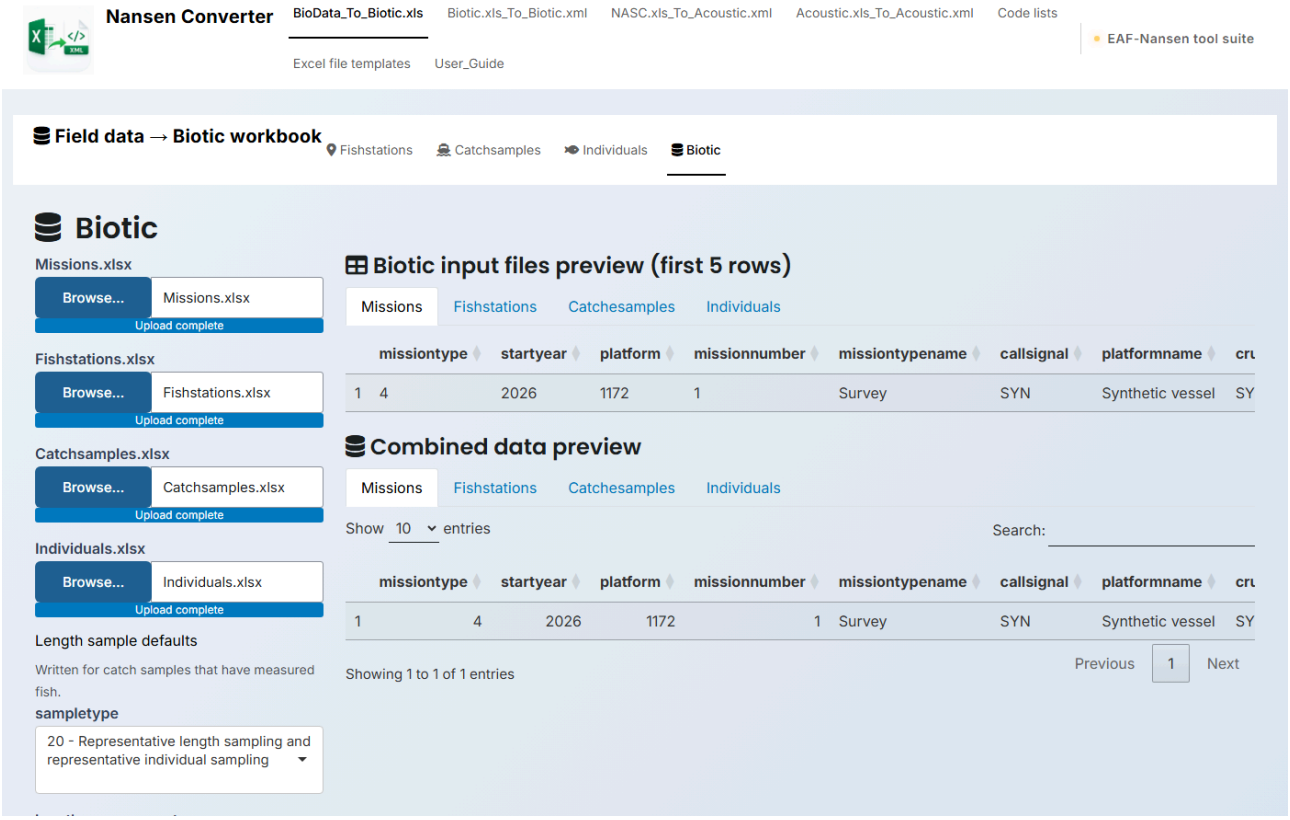


Figure 5: Biotic: the four files combined into the Biotic workbook

Import the four files you downloaded (Missions, Fishstations, Catchsamples, Individuals), check the **sample defaults** (`sampletype` 20, `lengthmeasurement` E, `sampleproducttype` 1, applied to the catch samples with measured fish) and click **Combine the data**. **Download Biotic** saves `Biotic.xlsx` with the four sheets `missions`, `fishstations`, `catchesamples` and `individuals`. The preview tabs show each sheet; you can also edit the workbook in Excel before the next step.

3 Biotic workbook → Biotic.xml

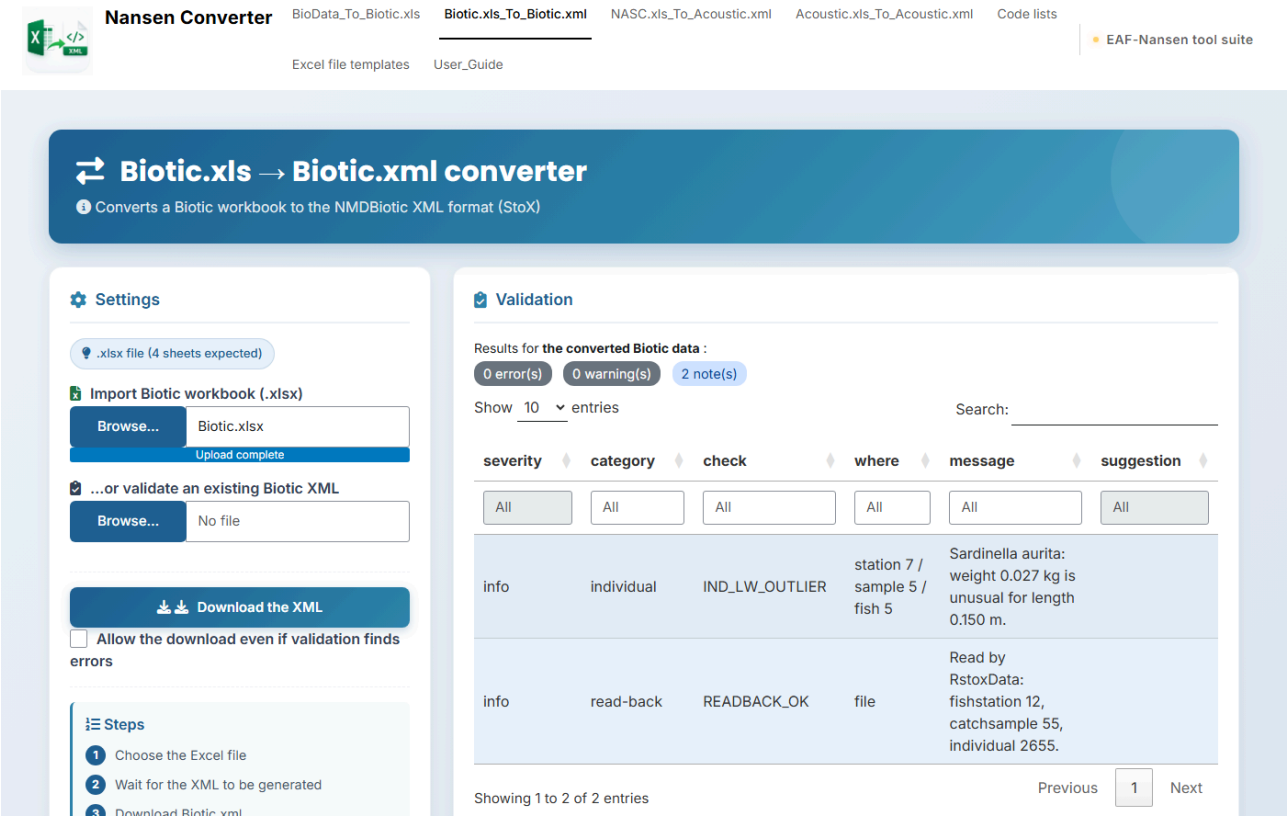


Figure 6: The Biotic XML tab with its Validation panel

Import `Biotic.xlsx` in the **Biotic.xls_To_Biotic.xml** tab. The XML is built and validated straight away (Figure 6):

- The **Validation** panel shows the number of errors, warnings and notes, and a table of the findings with where they are and what to do. Filter it by severity or category, or **Download findings (CSV)**.
- **Check unresolved names in WoRMS** looks up online any species name the taxonomy could not resolve.
- **Download the XML** saves `Biotic.xml`. It is blocked while there are errors; tick **Allow the download even if validation finds errors** to download anyway.
- **...or validate an existing Biotic XML** runs the same checks on an NMDBiotic file made elsewhere.

The *Generated XML preview* shows the start of the file. Errors are problems StoX would reject or that make the data wrong, such as broken links between tables or a missing aphia code. Warnings are values worth checking, such as a station on land or a fish far too heavy for its length. Notes are for information. All checks are listed in Section 9.

4 NASC → Acoustic workbook

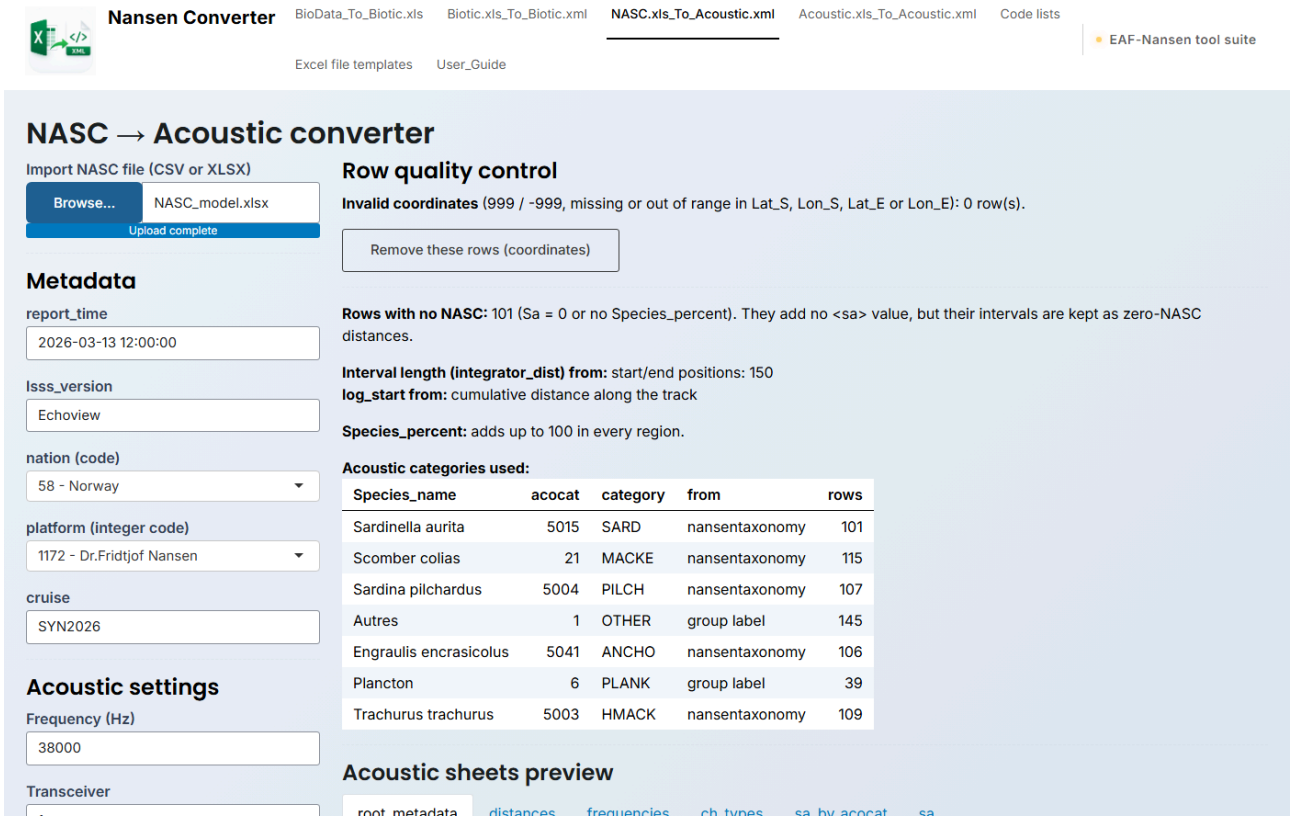


Figure 7: NASC: metadata and settings on the left, the quality checks and acoustic categories on the right

The **NASC.xls_To_Acoustic.xml** tab reads an Echoview export with one row per interval, layer, region and species (`.csv` or `.xlsx`). It needs these columns:

`Interval`, `Layer`, `Date_S`, `Time_S`, `Date_E`, `Time_E`, `Lat_S`, `Lon_S`, `Lat_E`, `Lon_E`, `PRC_NASC`, `Species_name`, `Species_percent`, `Exclude_above_line_depth_mean`, `Exclude_below_line_range_mean`, `Exclude_below_line_range_max`, `Exclude_below_line_depth_min`.

`Region_ID` is optional; with it the app checks that `Species_percent` adds up to 100 in every region. `VL_start`/`VL_end` (vessel log) or `Dist_S`/`Dist_E` (GPS distance), when present, give the interval lengths.

1. **Import** the NASC file.
2. Fill in the **Metadata**: `report_time`, `lsss_version`, `nation` and `platform` (codes from the code lists) and `cruise`.
3. Check the **Acoustic settings**: frequency (38000 Hz), transceiver (1), Sv threshold (−70 dB), pelagic channel thickness (10 m), upper interpretation depth (10 m), lower integration depth (500 m), and whether species get specific acoustic categories (for example PILCH, ANCHO) or only the broad ones (PEL1, PEL2, MESFI, BOTT, OTHER).
4. Click **Process file**, review the checks, then **Download Acoustic.xlsx**.

Each species' NASC is its share of the region, `PRC_NASC × Species_percent / 100`, summed by interval, acoustic category and channel. The category comes from nansentaxonomy, or from the group labels used in Echoview (for example *Autres* → OTHER, *Plancton* → PLANK). The **Row quality control** panel (Figure 7) reports:

- rows with invalid coordinates (999, −999, missing or out of range), which you can remove;
- rows with no NASC, whose intervals stay in the file as zero-NASC distances;
- where the interval lengths and `log_start` come from (vessel log, GPS distance, or the start and end positions);
- regions whose `Species_percent` does not add up to 100;
- the acoustic category given to each species, and where it came from.

`Acoustic.xlsx` has six sheets: `root_metadata`, `distances`, `frequencies`, `ch_types`, `sa_by_acocat` and `sa`. The preview tabs show each one.

5 Acoustic workbook → Acoustic XML

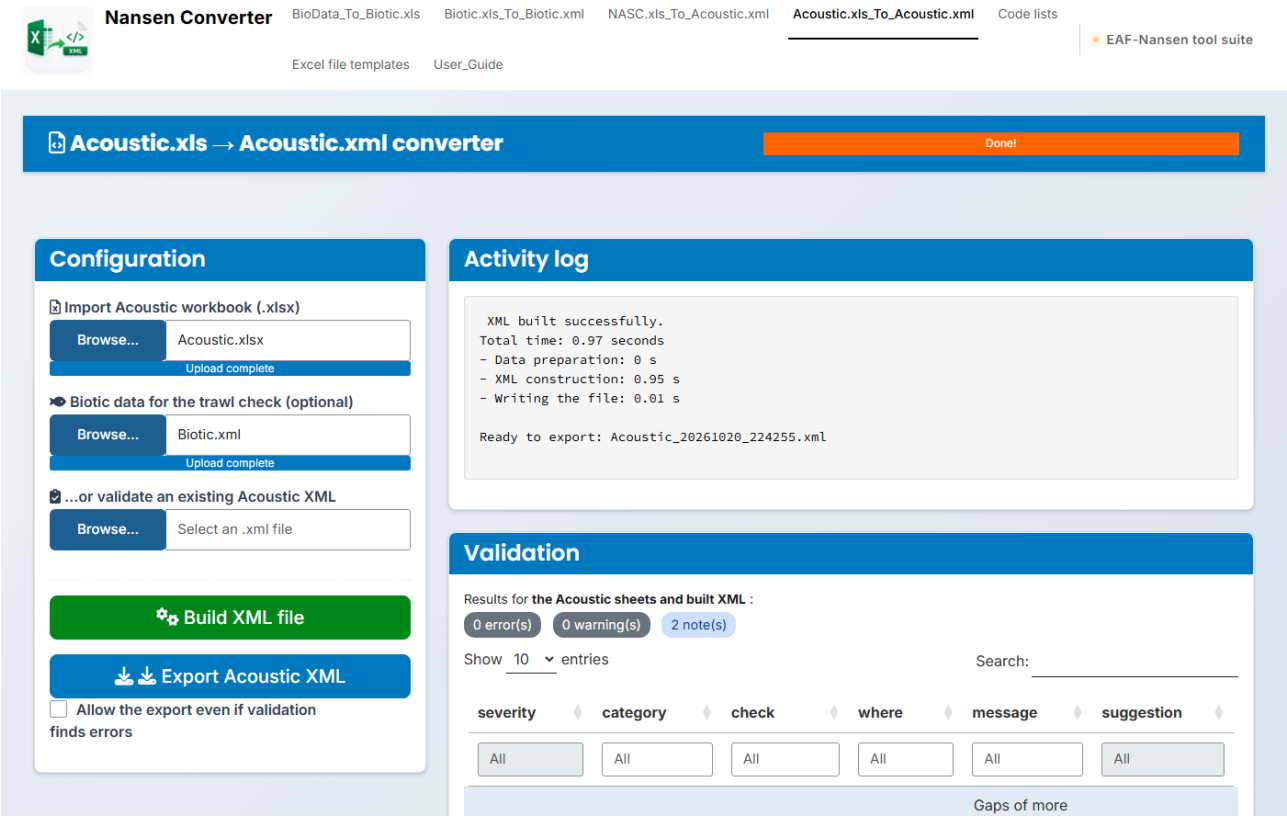


Figure 8: The Acoustic XML tab after building the XML

In the **Acoustic.xls_To_Acoustic.xml** tab:

1. **Import** `Acoustic.xlsx`.
2. Optionally add the **Biotic data for the trawl check** (`Biotic.xlsx` or `Biotic.xml`). The validation then reports acoustic categories with no trawl catch to support them, and trawls outside the acoustic period.
3. Click **Build XML file**. The *Activity log* shows the progress.
4. Review the **Validation** panel and **Export Acoustic XML**. As for Biotic, errors block the export unless you tick **Allow the export even if validation finds errors**.

...or validate an existing Acoustic XML checks an NMD Echosounder file made elsewhere.

6 Code lists

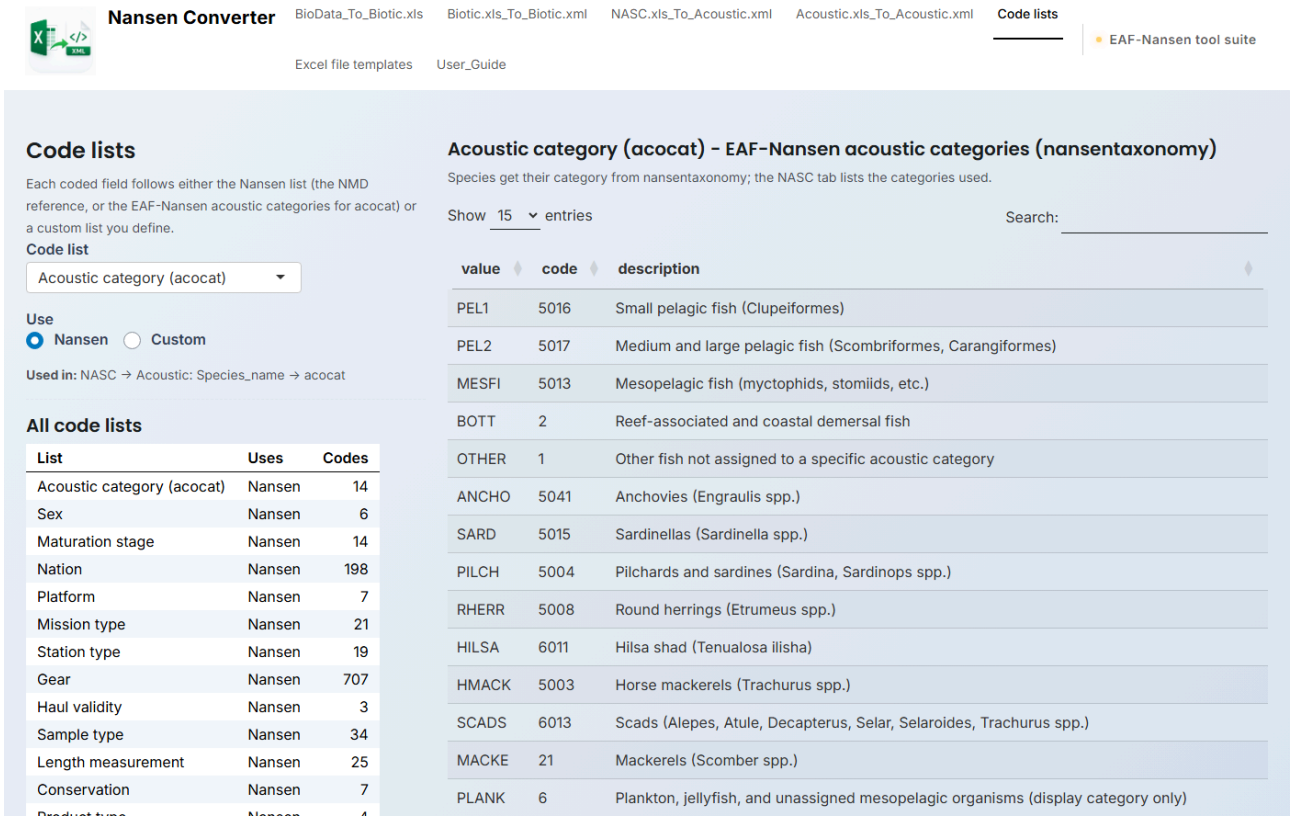


Figure 9: The Code lists tab

Every coded field uses one of the code lists below. For each list, choose under **Use**:

- **Nansen**: the NMD reference codes (from nansencore), or for acoustic categories the EAF-Nansen categories (from nansentaxonomy). This is the default.
- **Custom**: your own table. **Fill the custom list from the Nansen list** gives you a starting point; for acocat you can also start from the old LPA codes. Right-click the table to add or remove rows.

For the mapped lists (acocat, sex, maturation stage), `value` is what appears in your input file and `code` is what is written to the XML; a `value` of `*` gives the code for any value that is not listed.

Table 4: The code lists, with the number of codes in the Nansen version

List	Nansen list	Codes	Used in
Acoustic category (acocat)	EAF-Nansen acoustic categories	14	NASC → Acoustic: Species_name → acocat
Sex	NMD reference	6	Individuals: sexe (F / M / I) → sex
Maturation stage	NMD reference	14	Individuals: sms → maturationstage
Nation	NMD reference	198	Station defaults; NASC metadata
Platform	NMD reference	7	Mission metadata; NASC metadata
Mission type	NMD reference	21	Mission metadata
Station type	NMD reference	19	Station defaults
Gear	NMD reference	707	Station defaults
Haul validity	NMD reference	3	Station defaults
Sample type	NMD reference	34	Biotic sample defaults
Length measurement	NMD reference	25	Biotic sample defaults
Conservation	NMD reference	7	Catch sample defaults
Product type	NMD reference	4	Catch, sample and individual product type
Length resolution	NMD reference	13	Individuals

Export code lists (CSV) saves all lists and their settings; **Import code lists (CSV)** loads them back, for example on another computer. Your choices are also saved on this computer and reloaded the next time the app starts.

7 Excel file templates

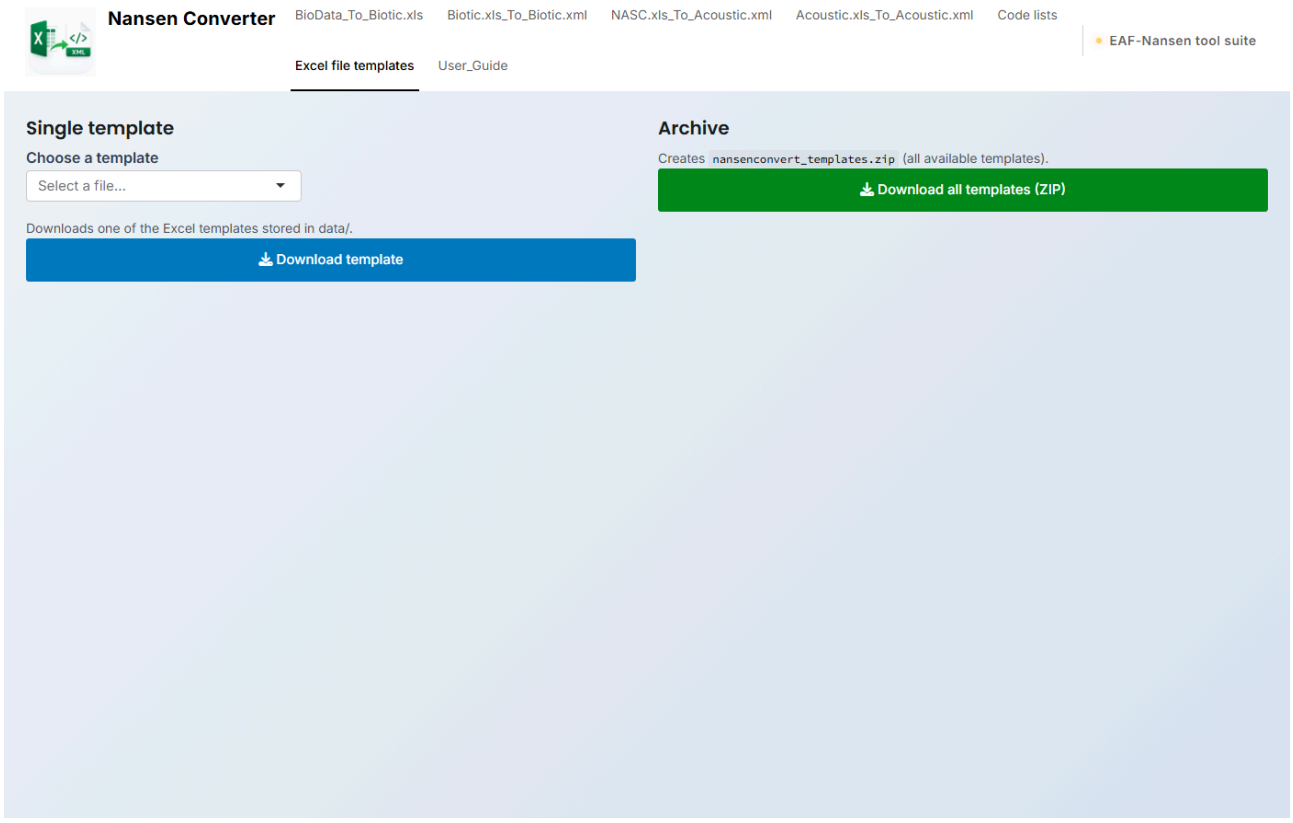


Figure 10: The Excel file templates tab

Download one template, or all of them as `nansenconvert_templates.zip`. They hold a fictional three-day survey (12 stations off northwest Africa, 10–12 March 2026), so you can try every converter without real data and see the expected columns:

- **Stations_model.xlsx**: Station, Date, Durée_chalutage, Cap_chalutage, Vitesse_chalutage, Longueur_funes, Heure_filage, Latitude_filage, Longitude_filage, Prof_fond_filage, Prof_filet_filage, Heure_virage, Latitude_virage, Longitude_virage, Prof_fond_virage, Prof_filet_virage, Note
- **Catches_model.xlsx**: Station, Espece, Ech_P, Tot_P, Tot_N
- **Length-frequency_model.xlsx**: station, espece, Poids_Mens, taille_cm, nombre
- **Biology_model.xlsx**: id_campagne, mois, annee, saison, navire, chef_mission, stock, type_mission, espece, station, latitude, longitude, nombre, poids_g, taille_cm, sexe, sms, graisse, aliment, couleur
- **NASC_model.xlsx**: Interval, Layer, Region_ID, PRC_NASC, Date_S, Time_S, Date_E, Time_E, Lat_S, Lon_S, Lat_E, Lon_E, Exclude_above_line_depth_mean, Species_name, Species_percent, Exclude_below_line_range_mean, Exclude_below_line_range_max, Exclude_below_line_depth_min

8 Common problems

"Missing columns" when processing a NASC file. Rename the columns of your export to the names in Section 4, or export them from Echoview.

A species keeps coming back as not found. Check its spelling in the species dialog or with **Check these names in WoRMS**. If it is a valid name that nansentaxonomy does not hold yet, download it from the dialog and send it to the nansentaxonomy maintainers.

The XML download is greyed out. Validation found errors. Read the *suggestion* column of the findings, fix the workbook and import it again. Download anyway only when you know the error does not matter for your use.

Wrong codes in the XML. Check the Code lists tab: a list set to *Custom* replaces the Nansen codes everywhere it is used.

Nation or area missing. They have no default; set them under *Station defaults* (Fishstations) and in the NASC metadata.

9 Validation checks

The Validation panels can report the checks below, grouped by the *category* shown in the findings table, with their severity. The thresholds behind them (plausible speeds, depths, lengths and weights) are in `helpers/validation.R`.

- **biology**: ACO_NO_TRAWL_SUPPORT (warning), ACO_TRAWL_OUTSIDE_PERIOD (note)
- **catch**: CATCH_NEGATIVE (error), CATCH_SUBSAMPLE_EXCEEDS (error), CATCH_MEAN_WEIGHT (warning), CATCH_WEIGHT_MISSING (warning), IND_EXCEEDS_SAMPLECOUNT (warning), CATCH_COUNT_MISSING (note), CATCH_DUPLICATE_SPECIES (note)
- **codes**: ACO_ACOCAT_INVALID (error), CODE_INVALID (warning)
- **depth**: ACO_CHANNEL_RANGE (warning)
- **distance**: ACO_COORD_RANGE (error), ACO_DIST_DUPLICATE (error), ACO_TIME_INVALID (error), ACO_TIME_ORDER (error), ACO_COORD_MISSING (warning), ACO_LENGTH_INVALID (warning), ACO_LOG_DECREASING (warning), ACO_ON_LAND (warning), ACO_POSITION_JUMP (warning), ACO_SPEED_RANGE (warning), ACO_TIME_OVERLAP (warning), ACO_TIME_ZERO (warning), ACO_TIME_GAP (note)
- **individual**: IND_LENGTH_RANGE (warning), IND_LENGTH_UNIT (warning), IND_WEIGHT_RANGE (warning), IND_LW_OUTLIER (note)
- **metadata**: ACO_META_MISSING (error), ACO_PLATFORM_INTEGER (error)
- **nasc**: ACO_NASC_MISSING (error), ACO_NASC_NEGATIVE (error), ACO_SPECIES_PERCENT (warning), ACO_NASC_OUTLIER (note)
- **read-back**: READBACK_COUNT (error), READBACK_FAILED (error), READBACK_NOT_RUN (note), READBACK_OK (note)
- **schema**: SCHEMA_INVALID (error), SCHEMA_UNREADABLE (error), SCHEMA_NOT_RUN (note)

- **station**: STN_COORD_RANGE (error), STN_COORD_MISSING (warning), STN_COORD_ZERO (warning), STN_DATE_ORDER (warning), STN_DATE_OUTSIDE_MISSION (warning), STN_DEPTH_RANGE (warning), STN_DISTANCE_POSITIONS (warning), STN_DISTANCE_RANGE (warning), STN_DUPLICATE_TIME (warning), STN_EFFORT_INCONSISTENT (warning), STN_FISHING_BELOW_BOTTOM (warning), STN_GEAR_GEOMETRY (warning), STN_ON_LAND (warning), STN_SOAKTIME_RANGE (warning), STN_SPEED_RANGE (warning)
- **structure**: CATCH_ID_INVALID (error), CATCH_KEY_DUPLICATE (error), CATCH_ORPHAN (error), IND_KEY_DUPLICATE (error), IND_SPECIMEN_INVALID (error), MISSION_KEY_INTEGER (error), MISSION_KEY_MISSING (error), MISSION_MISSING (error), STN_SERIAL_INVALID (error), XF_SERIAL_REUSED (error), IND_NOT_LINKED (warning), IND_ORPHAN (warning), IND_SPECIMEN_MISSING (warning), MISSION_MULTIPLE (warning)
- **taxonomy**: TAX_APHIA_MISSING (error), TAX_APHIA_INCONSISTENT (warning), TAX_UNRESOLVED (warning), TAX_NOT_SPECIES (note), TAX_PLACEHOLDER_TSN (note), TAX_SYNONYM (note)

10 Credits

Nansen Converter is part of the EAF-Nansen tool suite. It is based on the LPA **Excel XML Converter** (2026) by KMZ.